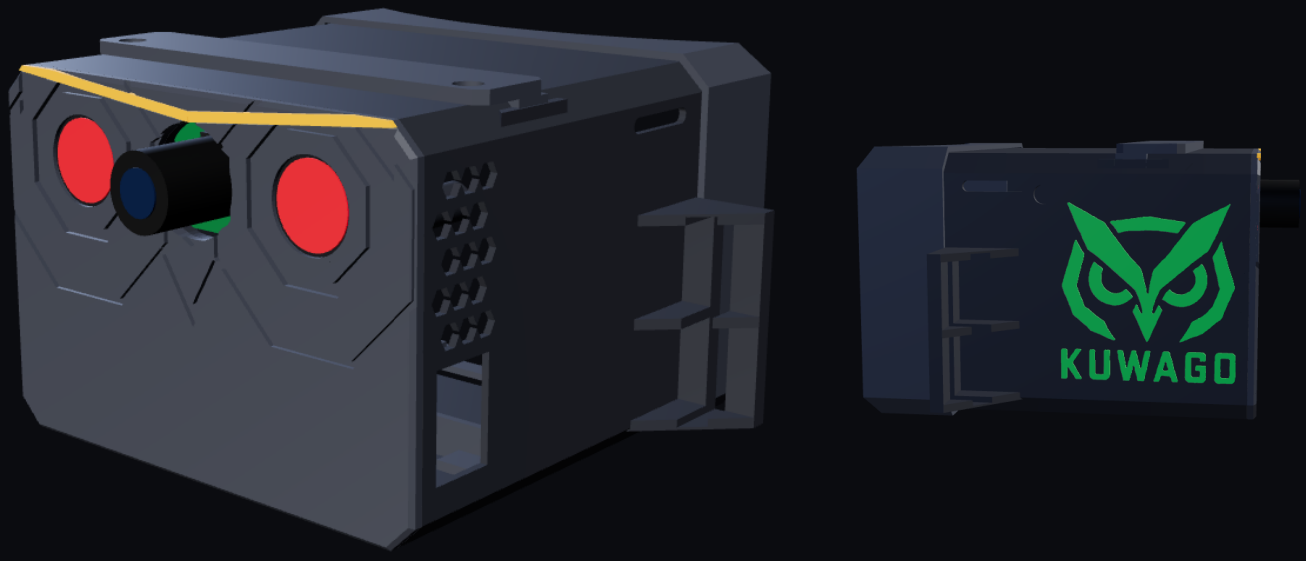
KUWAGO

# Homemade Night Vision Goggles

PRINT & ASSEMBLY GUIDE · v2.2 · 14 NOV 2026




Raspberry Pi Zero · IR night-vision camera · 3.5" HDMI screen · Google Cardboard lenses

"Kuwago" is Filipino for owl: the bird that sees in the dark.

Hoofie Puppet's Workshop

# What you need

Kuwago is a pair of digital night vision goggles. Two infrared LEDs light up the dark with light your eyes can't see, the camera can, and a Raspberry Pi Zero shows the live picture on a small screen behind two Google Cardboard lenses, one view for both eyes, like real PVS-7 goggles. The finished goggles measure about **97 × 69 × 87 mm**, and the full plastic set is about **139 g of PLA**.

## Electronics

| Part | Qty | Notes |
|---|---|---|
| Raspberry Pi Zero (or Zero 2 W for a smoother picture) | 1 | The brain. Same size and holes either way. |
| Pimoroni LiPo SHIM | 1 | Boosts the battery to 5 V, has the low-battery pin. **It does not charge the battery.** |
| IR night-vision camera, OV5647, with 2 IR LEDs | 1 | Board 73 × 25 mm, LED holders Ø18.5 × 13 mm. |
| Pi Zero camera ribbon (narrow end + wide end) | 1 | 10–15 cm is tidiest; a 30 cm one folds away fine. |
| 3.5" HDMI screen, MPI3508 (480 × 320) | 1 | Must be **HDMI**, not the slow SPI type. |
| Mini-HDMI cable, 90° down-angle head, FPV flat-ribbon type | 1 | Only ~7 mm under the Pi for the plug. |
| LiPo 3.7 V 2000 mAh (PKCELL LP803860) | 1 | Lives in the battery pod. |
| TP4056 USB-C LiPo charger, protected version | 1 | Charges the battery inside the pod. |
| Mini rocker switch (19 × 13 mm cut-out) | 1 | Wired to SHIM EN + GND. |
| JST-PH 2-pin extension lead, ~40 cm | 1 | Battery pod to goggles, along the strap. |
| MicroSD card 8 GB+, thin hookup wire | – | Raspberry Pi OS Lite; screen power and switch wires. |
| *Optional:* MAX17048 LiPo fuel gauge | 1 | For an on-screen battery % bar. |

## Optics and hardware

| Part | Qty | Notes |
|---|---|---|
| Google Cardboard bi-convex lenses, Ø25 mm, 40–45 mm focal length | 2 | Edge 2.0–3.2 mm all fit. |
| M2.5 screws, 5–10 mm | ~16 | Pi 4 × 5 mm · face 4 × 5 mm · camera retainer 2 × 8 mm · tray cap 2 × 6 mm · pod lid 2 × 6 mm · thumbscrews 2 × 8–10 mm. |
| M2.5 nuts | 2 | For the lens-plate thumbscrews. |
| Elastic head strap, up to 38 mm wide | 1 | An old goggle or headlamp strap works. |
| Foam, 3–5 mm (EVA or a VR face cushion) | 1 | Face seal; a scrap for the pod's back. |
| 550 paracord, about 55 cm | 1 | Gutted, as a solid flexible sleeve for the battery lead. |
| CA glue, painter's or masking tape, small screwdriver, soldering iron | – | Tape is for the logo transfer trick. |

> **TIP** Don't print the screen tray until your screen arrives. MPI3508 clones vary in size, so measure yours (board outline, screen area, where the HDMI port and pins are) and get the tray regenerated to fit.

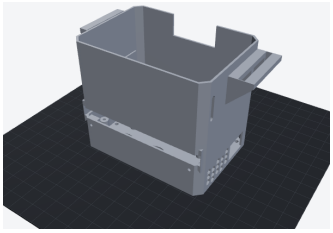
# Print settings

Tuned for a **Creality Ender 5 Max** on its PEI plate with a high-flow (Unicorn / Hi-format) nozzle. Every part is already oriented in its STL: just drop it on the bed. **No supports anywhere.**

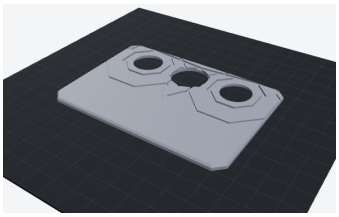
| Setting | Value | Why |
|---|---|---|
| Material | PLA, matte black (body) · green (logo) · translucent amber (brow, optional) | |
| Nozzle / bed | 210–215 °C / 60 °C | Hotter end of PLA suits the high-flow nozzle. |
| Bed prep | Clean PEI, thin coat of Aqua Net | Big flat parts (shell front, face) must not lift. |
| Layer height | 0.20 mm (0.12 mm for small parts, see below) | |
| Walls / top / bottom | 3 walls · 5 top · 4 bottom | Most walls are 1.4 mm: 3 lines makes them solid. |
| Infill | 20 % gyroid | Only the thicker spots have any infill. |
| Supports | **None** | All overhangs are 45° or bridges under 20 mm. |
| Cooling | 100 % after layer 2 | Clean 45° chamfers and bridges. |
| Speed | Normal profile; 30 mm/s first layer | Slow down to 40 mm/s for the logo inlays. |
| Brim | None (5 mm brim on the shell if your bed is cold or drafty) | The shell is 85 mm tall. |
| Elephant's foot | Compensation 0.15 mm | Keeps the flush owl face and inlay recesses crisp. |

**NOTE** Your E5M doesn't like mid-print colour changes, so nothing here needs one: the green logo is 12 separate inlay pieces printed on their own plate.

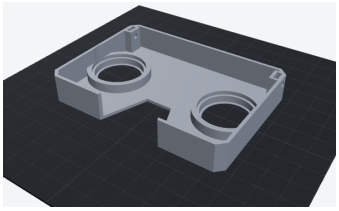
# Part by part



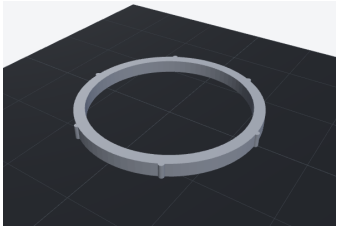
**Shell** ×1 · kw-01-shell.stl · ≈57.8 g
**Orientation:** Front wall down on the bed (the open back faces up).
**Settings:** 0.20 mm · 3 walls · 20 %
Biggest print, 129 × 69 × 85 mm. Optional 5 mm brim.



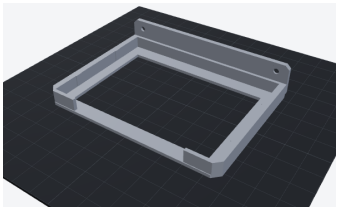
**Owl face** ×1 · kw-02-owl-face.stl · ≈13.7 g
**Orientation:** Flat back down, engraved owl face up.
**Settings:** 0.20 mm · 3 walls · 20 %
Turn ironing on for the top surface for a clean face.



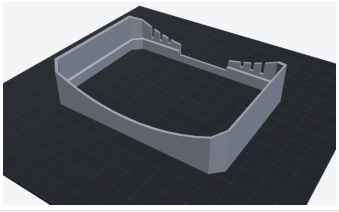
**Lens / focus plate** ×1 · kw-03-lens-plate.stl · ≈17 g
**Orientation:** Eye side down.
**Settings:** 0.20 mm · 3 walls · 20 %
Nut pockets open at the top: drop the nuts in after.



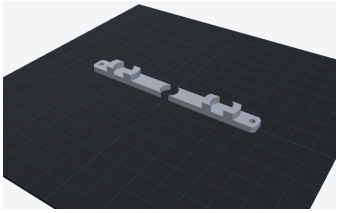
**Lens rings** ×2 · kw-04-lens-ring.stl · ≈0.3 g
**Orientation:** Flat.
**Settings:** 0.12 mm · 3 walls · 100 %
Tiny grip ribs make the press fit.



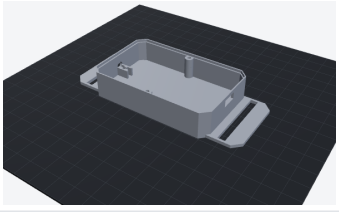
**Screen tray (v0)** ×1 · kw-05-screen-tray-mpi3508.stl · ≈12.2 g
**Orientation:** Screen frame down.
**Settings:** 0.20 mm · 3 walls · 20 %
Wait for your screen; it gets regenerated to fit.



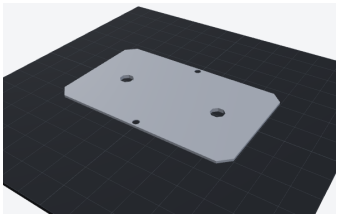
**Face shroud** ×1 · kw-06-face-shroud.stl · ≈9.3 g
**Orientation:** Sleeve end down, curved face edge up.
**Settings:** 0.20 mm · 3 walls · 20 %
Hidden anti-fog vents by the nose.



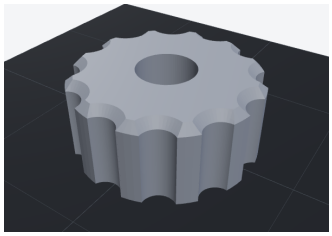
**Camera retainer** ×1 · kw-07-camera-retainer.stl · ≈2.4 g
**Orientation:** Flat bar down, pads up.
**Settings:** 0.20 mm · 3 walls · 30 %



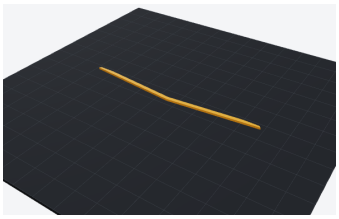
**Battery pod** ×1 · kw-08-battery-pod.stl · ≈17 g
**Orientation:** Floor down.
**Settings:** 0.20 mm · 3 walls · 20 %
Strap slots at both ends; strain relief inside.



**Pod lid** ×1 · kw-09-battery-pod-lid.stl · ≈6.1 g
**Orientation:** Outer face down.
**Settings:** 0.20 mm · 3 walls · 100 %



**Thumb knobs** ×2 · kw-10-thumb-knob.stl · ≈0.8 g
**Orientation:** Flat, screw-head pocket up.
**Settings:** 0.12 mm · 4 walls · 100 %
Press an M2.5 screw head into each.



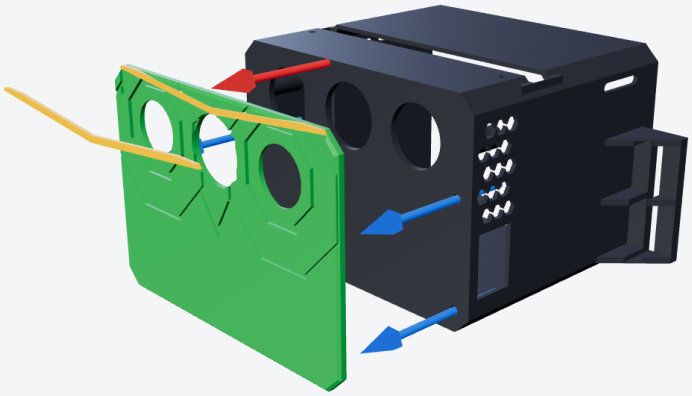
**Brow accent (optional)** ×1 · kw-11-brow-accent.stl · ≈0.2 g
**Orientation:** Flat.
**Settings:** 0.12 mm · 100 % · translucent amber
One V-shaped strip; presses flush into the brow groove.




**Logo inlays** ×1 plate · kw-12-logo-inlays.stl · ≈0.9 g
**Orientation:** Flat, already laid out like the logo.
**Settings:** 0.12 mm · 100 % · green · 40 mm/s
12 pieces: owl head, 2 cheeks, 2 eyes, brow bar, K-U-W-A-G-O.

# Assembly

In each picture the part you are adding is **green** (or amber), and **red** / **blue** arrows show where it goes. See-through grey parts are already fitted.



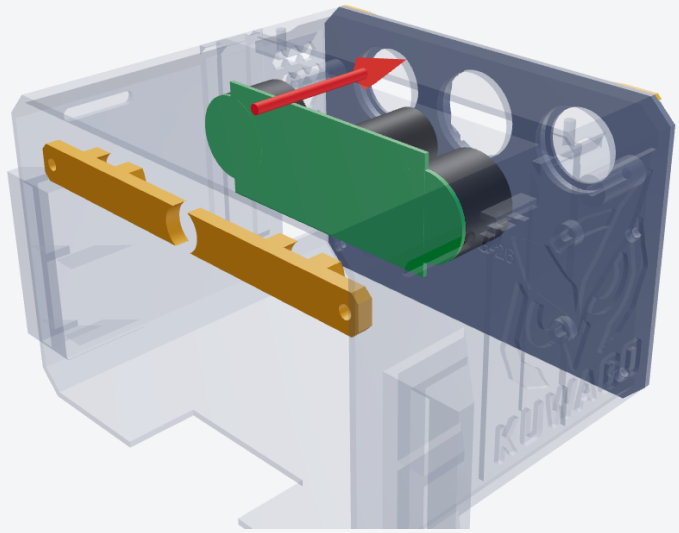
## 1 · Owl face onto the shell

- Place the owl face on the front of the shell.
- Fix it from the **inside** with 4 × M2.5 × 5 mm screws (blue arrows).
- Optional: press the amber brow accent into the brow groove (red arrow). A drop of CA if it's loose.



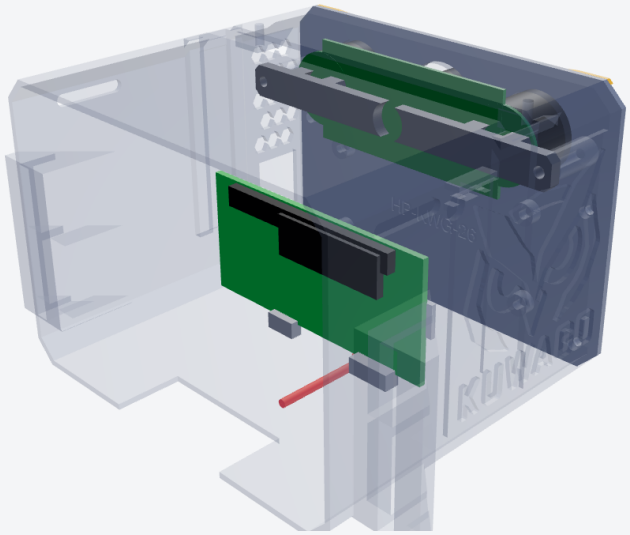

## 2 · Logo inlays on the left side

- **Transfer trick:** while the 12 green pieces are still on the bed, press painter's tape over all of them and peel. They lift in the right positions.
- Put a tiny dot of CA in each recess, press the whole sheet in, then peel the tape away.
- Or place them one by one with tweezers: owl head, cheeks, eyes, brow bar, then the letters.



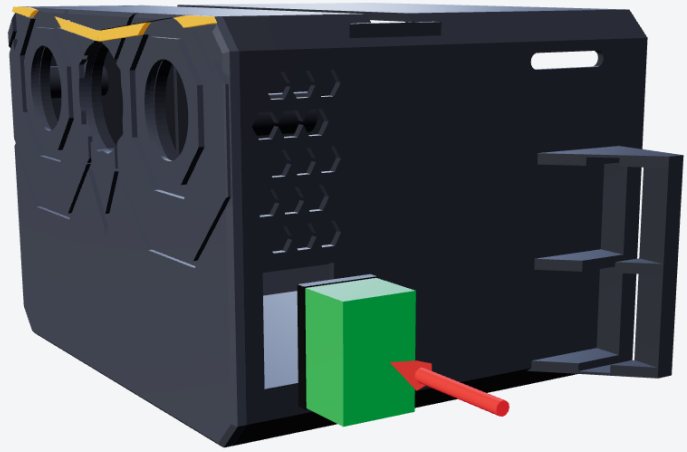
## 3 · Camera board and retainer

- From the back, push the camera board forward so the two IR LED holders go through the LED holes and the lens through the beak.
- The LED rims stop on a small ledge at the face, so they sit flush.
- Screw the camera retainer (amber) across the back of the board onto the two posts: 2 × M2.5 × 8 mm.
- The board's light sensors end up in the dark, so the IR stays on whenever the goggles run. That's what you want.



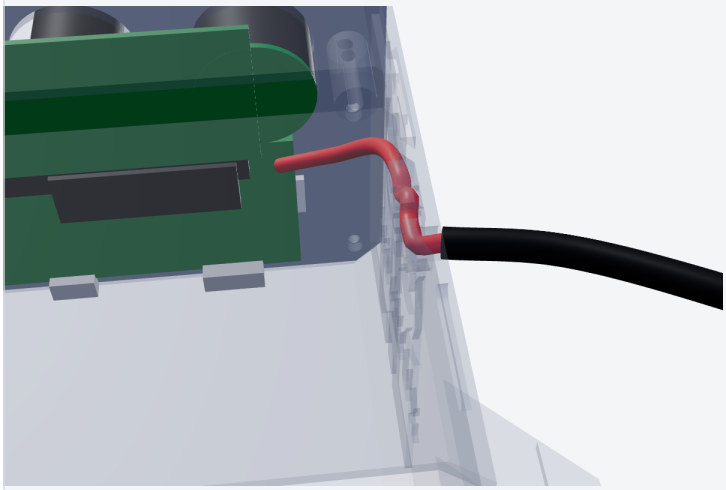
## 4 · Pi Zero + LiPo SHIM

- Plug the 90° mini-HDMI head and the camera ribbon into the Pi **first**.
- Mount the Pi under the camera on the 4 standoffs (4 × M2.5 × 5 mm): components facing you, ports facing down, microSD on the left.
- Fold any ribbon slack flat beside the Pi.



## 5 · Power switch

- Snap the rocker switch into the cut-out on the right side, standing upright.
- Wire it to the SHIM's **EN** and **GND** pads: switch closed = goggles off. Never put it inline on the battery.



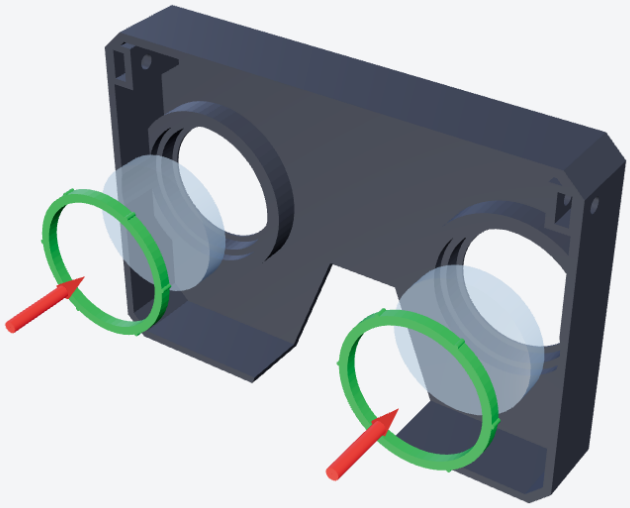
## 6 · Battery lead and strain relief

- Sleeve the lead in gutted 550 paracord first (see *Protecting the battery lead*).
- Feed the bare lead in through the hole on the left side and push the paracord's sealed end into the **4.2 mm socket** until it stops (about 3.5 mm in). It works as a built-in grommet.
- Press the bare lead into the **zig-zag channel** just inside so it snakes between the bumps.
- Leave a little slack, then plug it into the SHIM. A tug on the cable now stops at the zig-zag, not at the plug.



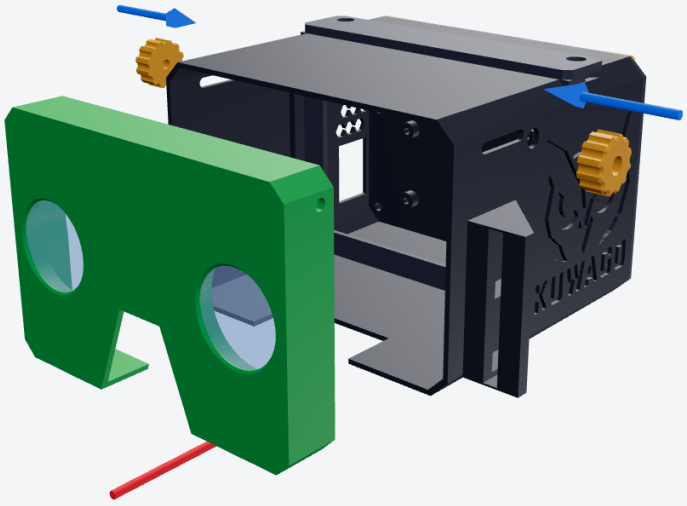
## 7 · Screen tray

- Press the screen into the tray, screen facing the window.
- Slide the tray down through the top slot. Fix the cap with 2 × M2.5 × 6 mm.
- Connect HDMI and the 5 V / GND wires from the SHIM behind it.
- **Swapping screens later:** unplug HDMI + power, undo 2 screws, lift out. Nothing else moves.



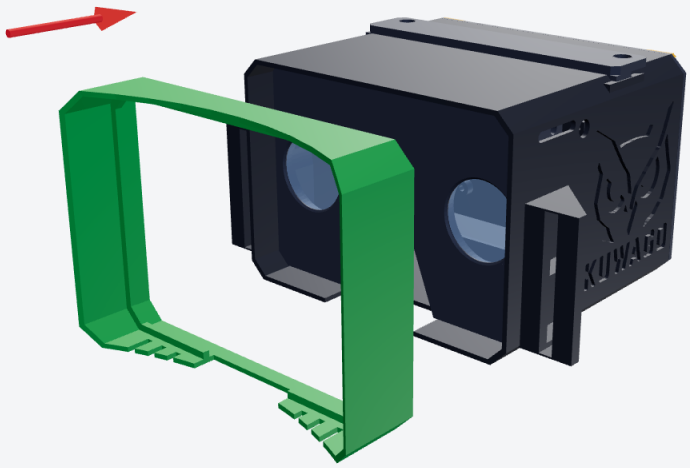
## 8 · Lenses into the lens plate

- Drop each Cardboard lens into its seat from the screen side.
- Press a lens ring on top until it just touches the lens. It holds by friction, whatever the lens edge thickness.



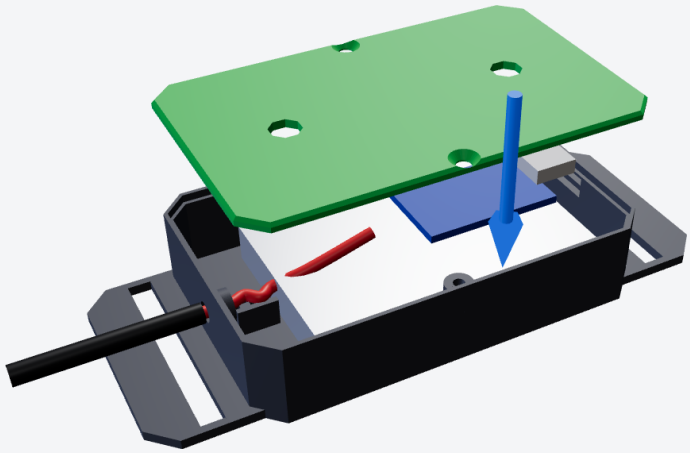

## 9 · Lens plate and thumb knobs

- Drop an M2.5 nut into each side pocket of the lens plate.
- Slide the plate into the back of the shell, lenses toward you.
- Press an M2.5 × 8–10 mm screw head into each thumb knob and thread it through the side slots into the nuts.



## 10 · Face shroud

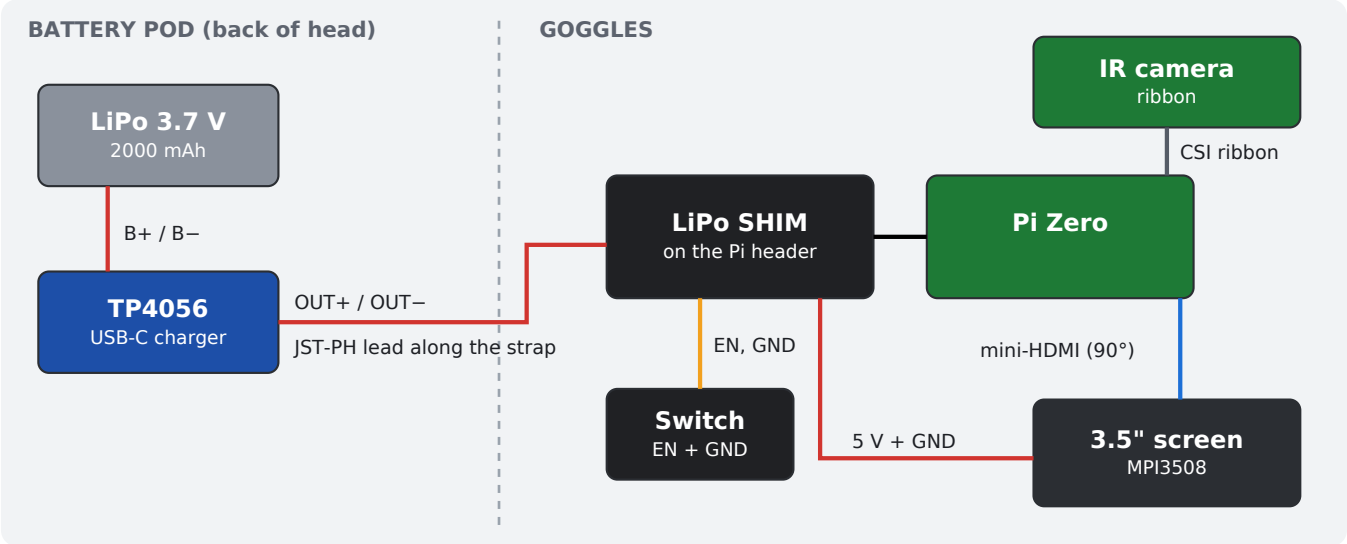
- Glue foam to the curved face edge of the shroud.
- Push the shroud over the back of the shell until it seats on the rim.
- Thread the head strap through the slots in the two side wings.



## 11 · Battery pod

- Battery on the floor of the pod. TP4056 charger on top, its USB-C port at the opening.
- Push the paracord's sealed end into the pod's 4.2 mm end socket and press the bare lead into the zig-zag channel on the floor.
- Fit the lid (2 × M2.5 × 6 mm). Thread the strap through the pod's end slots so it sits at the back of your head, lid facing out.
- Stick a scrap of foam on the base where it touches your head.

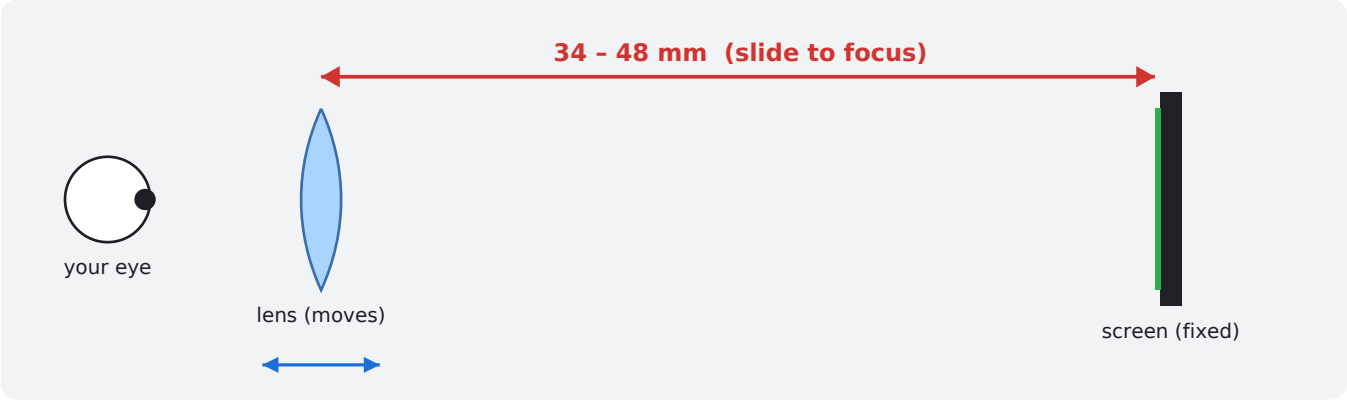
# Wiring




- **Battery → TP4056:** battery wires to B+ / B−. The TP4056's OUT+ / OUT− go to the extension lead.
- **Lead → SHIM:** the JST-PH lead runs along the strap into the goggles' left side and plugs into the SHIM.
- **Switch:** between the SHIM's EN and GND. Closed = off, open = on.
- **Screen power:** two wires from the SHIM / Pi header 5 V and GND to the screen's 5 V and GND pins.
- **Video and camera:** 90° mini-HDMI from the Pi to the screen; camera ribbon from the camera board to the Pi.
- **Optional fuel gauge (MAX17048):** on the battery line, 4 wires to the Pi: 3V3, GND, SDA, SCL. Tape it to the inside wall.

**SAFETY** Charge only with the TP4056 in the pod, never through the SHIM. Unplug the lead before any soldering.

# Setting the focus




- Turn the goggles on and loosen both thumb knobs.
- Slide the lens plate forward or back until the picture is sharp and your eyes feel relaxed, like looking at something far away. Tighten both knobs.
- The 34–48 mm range covers both focal-length figures on the lens listing (40 and 45 mm).
- Eye spacing is set in software: the two pictures slide closer together or further apart until they line up for your eyes.

> **TIP** Start with 5–10 minute sessions and slow head turns. Because the camera is fixed to your head and both eyes see the same picture, most people get used to digital night vision within a few tries.

## Protecting the battery lead

**Recommended: gutted 550 paracord.** It's a tight woven tube, solid but very flexible, and both cable holes are 4.2 mm sockets made to take its end like a grommet.

- Cut about **55 cm** of 550 paracord (about 4 mm) and pull out all the white inner strands.
- Twist the two wires of the lead together lightly. Tape the end to a bobby pin and feed it through the empty sheath like a drawstring, **before** the second JST plug goes on.
- Push the sheath back on itself a little as you go: the weave gets denser and tougher.
- Melt each cut end quickly with a lighter so it can't fray. Then push the sealed ends into the sockets on the goggles and the pod.
- Reflective-tracer paracord shines silver in torch light. Use plain black if you want the goggles stealthy.

Other options, if you don't have paracord:

- **Braided cable sleeve** (expandable PET, 3 mm, Shopee / Lazada): neatest look. Finish its ends with short heat-shrink just outside each hole.
- **Aquarium airline tubing** (4 mm inside): the cheapest; thread the wire through before fitting the second connector.
- **Spiral wrap**: goes on after everything is assembled.
- **Silicone-jacketed 2-core wire** (24 AWG): tough and soft enough to need no sleeve at all.
- **Fabric (harness) tape**: the cloth tape used on car wiring.

Fix the lead to the strap with 3–4 small Velcro ties so it can't snag.

## Key dimensions (for remixes)

| What | Size |
|---|---|
| Goggles, outside | ≈ 97 × 69 × 87 mm (129 mm wide across the strap wings) |
| Lens centres | 63 mm apart, on the screen centre line |
| Lens seat | 25.3 mm bore, 1.0 mm lip with 23.0 mm opening, 27.8 mm ring counterbore |
| Camera | Board 6 mm+ behind the front wall; IR LEDs 27.25 mm either side of the lens |
| Pi Zero holes | 58 × 23 mm, M2.5 self-tapping into the standoffs |
| Screen tray | 92.4 mm wide, 8.2 mm channel, slides in from the top |
| Logo inlays | 46 mm tall logo, 1.0 mm pieces, recesses +0.10 mm per side and 1.06 mm deep |
| Battery pod | 80 × 50 × 17 mm (116 mm with strap tabs) |
| Cable sockets | 4.2 mm × 3.5–4 mm deep paracord grommets, 3.0 mm exit lip, zig-zag strain relief behind |